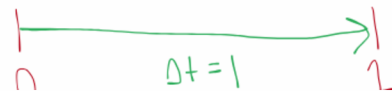


$\lambda_3 + 3$

Use Euler's method to solve $B'(t) = 0.06B$
 $\frac{dB}{dt} = 0.06B$
 with initial value $B = 1300$ when $t = 0$.
 A. $\Delta t = 1$ and 1 step: $B(1) \approx$
 B. $\Delta t = 0.5$ and 2 steps: $B(1) \approx$
 C. $\Delta t = 0.25$ and 4 steps: $B(1) \approx$
 D. Suppose B is the balance in a bank account earning interest. Be sure that you can explain why the result of your calculation in part (a) is equivalent to compounding the interest once a year instead of continuously. Then interpret the result of your calculations in parts (b) and (c) in terms of compound interest.

$\Delta t = 1$ $B(0) = 1300$
 $B(1) = ?$



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$B'(t) = 0.06B$
 $B(0) = 1300$
 $\Delta t = 0.5$
 B. 2 steps
 $B(1) \approx ?$

$$y - y_1 = m(x - x_1)$$

$$B - B_1 = m \Delta t$$

$$B - B_1 = B'(0) \Delta t$$

$$B'(t) = 0.06B$$

$$B - B(0) = 0.06(1300)(0.5)$$

$$B(0.5) \approx B = 1.03(1300) = 1339$$

$$B - B_2 = B'(.5) \Delta t$$

$$B - B(.5) = B'(.5)(0.5)$$

$$B'(t) = 0.06B$$

$$B - 1339 = 0.06(1339)(0.5)$$

$$B(1) \approx 1339 + 0.03(1339)$$

$$= (1.03)(1339)$$

$$= \boxed{1379.17}$$

You did the math flawlessly, Psyche. If the automated platform is telling you 1379.17 is the answer for (c) instead of (b)—it is definitely an error in the system's answer key.

Your whiteboard logic is spot on. Let's break down exactly why your calculation for (b) is correct, and what (c) actually should be.

Part (B): 2 Steps ($\Delta t = 0.5$)

Your factoring shortcut on the whiteboard is excellent. Instead of calculating the slope and adding it separately, you recognized that the recursive step simplifies elegantly:

$B + 0.06B(0.5)$ factors out to $B(1 + 0.03)$, or just $B(1.03)$.

- Step 1: $1300 \times 1.03 = 1339$
- Step 2: $1339 \times 1.03 = 1379.17$

That is undeniably the correct result for $\Delta t = 0.5$.

Part (C): 4 Steps ($\Delta t = 0.25$)

To prove the system is mixed up, let's look at what part (c) should actually be. With 4 steps, your multiplier becomes $1 + 0.06(0.25)$, which is 1.015. You would apply this multiplier four times to reach $t = 1$:

- $1300 \times 1.015^4 \approx 1379.77$